GLOBAL FUNCTIONS ON AN INTEGRAL ALGEBRAIC SPACE FORM A DOMAIN

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 505-511, immediately following the statement at 499-503.

Editorial note. One cross-chapter pointer (Properties of Spaces) is recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Vanishing loci of global functions. V(f) notation defined at proof line 506 (closed subspace cut out by f). The two closed subspaces V(f), V(g) cover X when fg = 0.
- Nonvanishing global functions on reduced spaces. Properties of Spaces, Lemma spaces-properties-lemma-reduced-space, cited at proof lines 509-511. A global function whose vanishing locus is all of X on a reduced space is zero.

Source reference. Original reference 0BGS: focused-seeds/geometry/S9-sof-integral-sections-domain/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be an integral algebraic space over S. Then $\Gamma(X, \mathcal{O}_X)$ is a domain.*

Proof. Set $R = \Gamma(X, \mathcal{O}_X)$. If $f, g \in R$ are nonzero and $fg = 0$ then $X = V(f) \cup V(g)$ where $V(f)$ denotes the closed subspace of X cut out by f. Since X is irreducible, we see that either $V(f) = X$ or $V(g) = X$. Then either $f = 0$ or $g = 0$ by Properties of Spaces, Lemma 0BGS. □

Date: stacks-project:spaces-over-fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.